# Detector-Plane Imaging (DPI):

# An Instrumentation-Grounded Measurement-Plane Framework


## Abstract

Detector-Plane Imaging (DPI) provides an instrumentation-grounded, detector-plane model of quantum observation, establishing that interference is encoded in a pre-detection coherence imprint and that detector-plane processing determines whether this structure is preserved or irreversibly lost. The QMCTB-01 benchmark is defined as a theoretical detector-plane causality test using a near-field double-slit configuration with identical excitation and propagation, evaluated under two detector regimes: intensity-only and correlation-preserving sensing.

By introducing fringe visibility (V) as a measurable criterion, DPI establishes a falsifiable distinction between detector architectures: intensity-only detection yields no stable interference (V ≈ 0), while correlation-preserving detection yields high-contrast interference (V ≈ 1). This framework treats quantum measurement as a hardware-defined selection process governed by coherence preservation at the sensor layer.

## 1. Introduction

Quantum interference experiments are traditionally interpreted through wave—particle duality, leading to ambiguity regarding when and how interference arises. Detector-Plane Imaging resolves this ambiguity by separating:

- Pre-detection coherence (imprint domain): a continuous, phase-encoded field structure

- Post-detection measurement outcome: a discretized, detector-dependent representation

Within this framework, particles remain primary excitations of quantum fields, while wave-like behavior emerges as a contextual imprint of propagation. Measurement does not create interference; it determines whether pre-existing coherence becomes observable.

## 2. Causality: Quantum Measurement Stack (QMS)

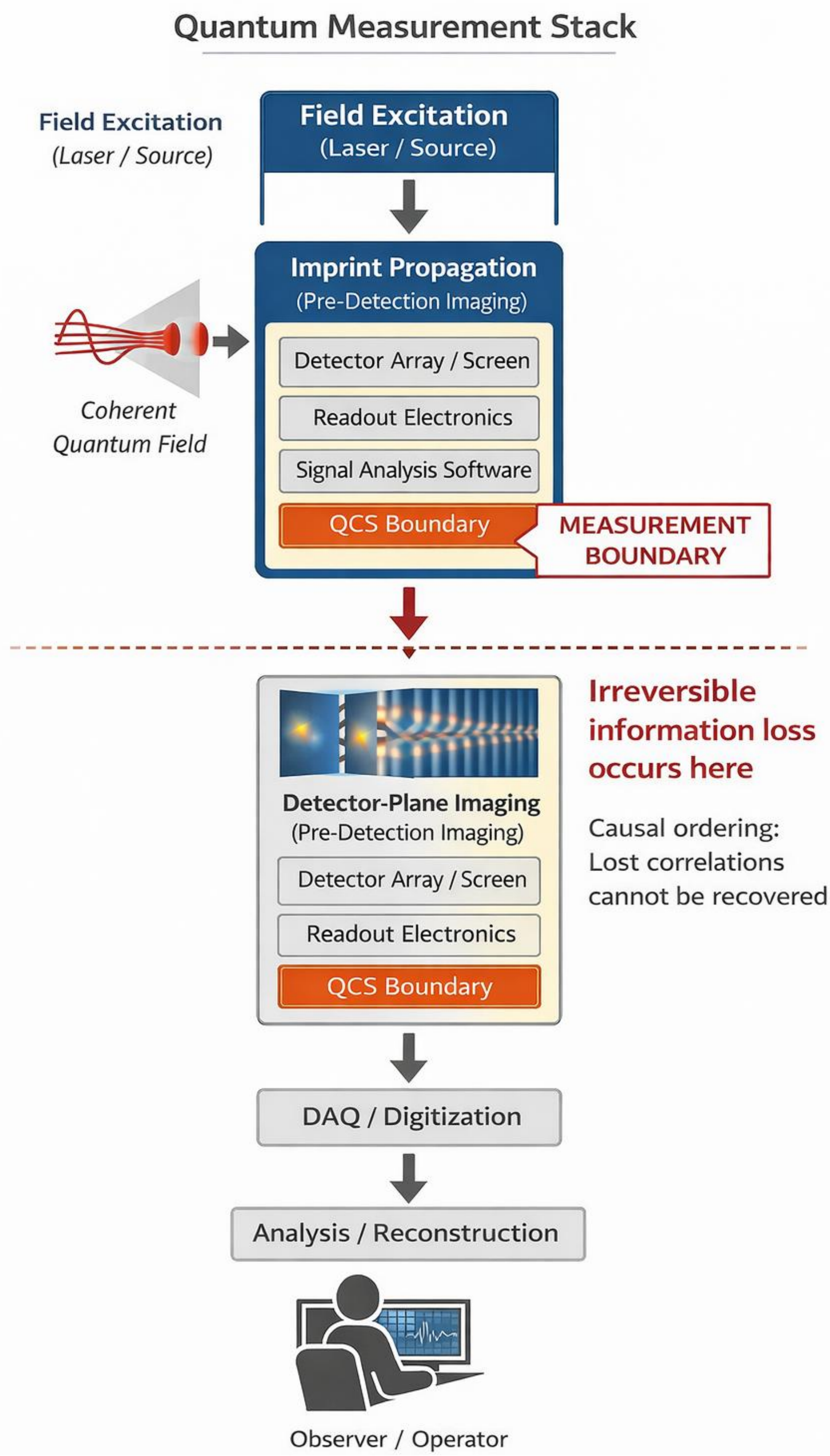
Field excitation → Imprint propagation → Detector plane → Readout → Analysis

The detector plane defines the measurement boundary where:

- Phase and correlation information may be preserved or discarded

- Information loss becomes irreversible

Once correlations are discarded at this stage, they cannot be recovered by downstream processing.


Quantum Measurement Stack
Field Excitation
(Laser / Source)
Field Excitation
(Laser / Source)
Imprint Propagation
(Pre-Detection Imaging)
Coherent
Quantum Field
Detector Array / Screen
Readout Electronics
Signal Analysis Software
QCS Boundary
MEASUREMENT
BOUNDARY
Irreversible
information loss
occurs here
Detector-Plane Imaging
(Pre-Detection Imaging)
Causal ordering:
Lost correlations
cannot be recovered
Detector Array / Screen
Readout Electronics
QCS Boundary
DAQ / Digitization
Analysis / Reconstruction
Observer / Operator

## 3. Mechanism: Detector–Plane Selection (DPI)

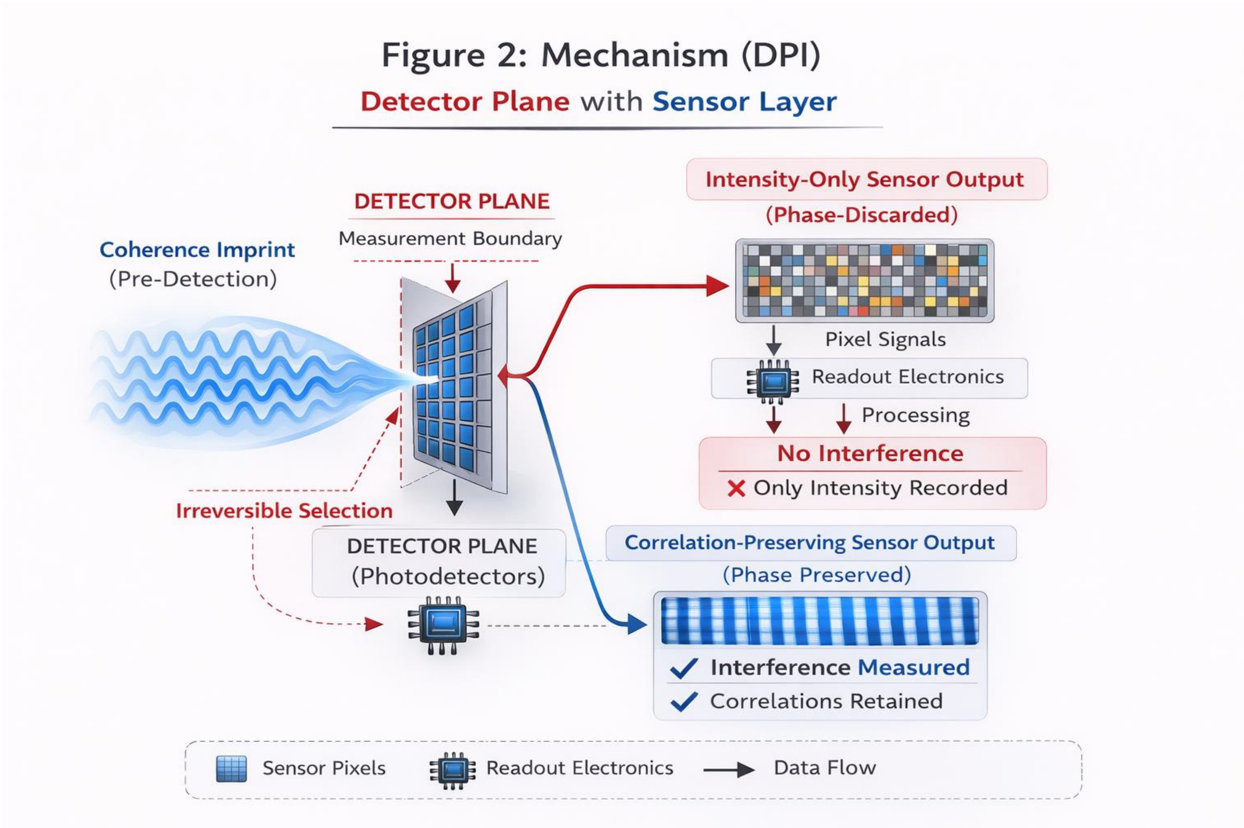
A single coherence imprint arrives at the detector as a continuous, phase–encoded structure. The detector determines whether the encoded correlations are preserved.

### Intensity–Only Detection

- Measures amplitude only ($|\psi|^2$)
- Discards phase correlations
- Output: No stable interference ($V \approx 0$)

### Correlation–Preserving Detection

- Retains phase relationships across the detector plane
- Preserves coherence structure
- Output: High–contrast interference ($V \approx 1$)




## 4. Validation: QMCTB–01 Benchmark

Identical excitation, propagation, and photon statistics are maintained. Only detector architecture changes.

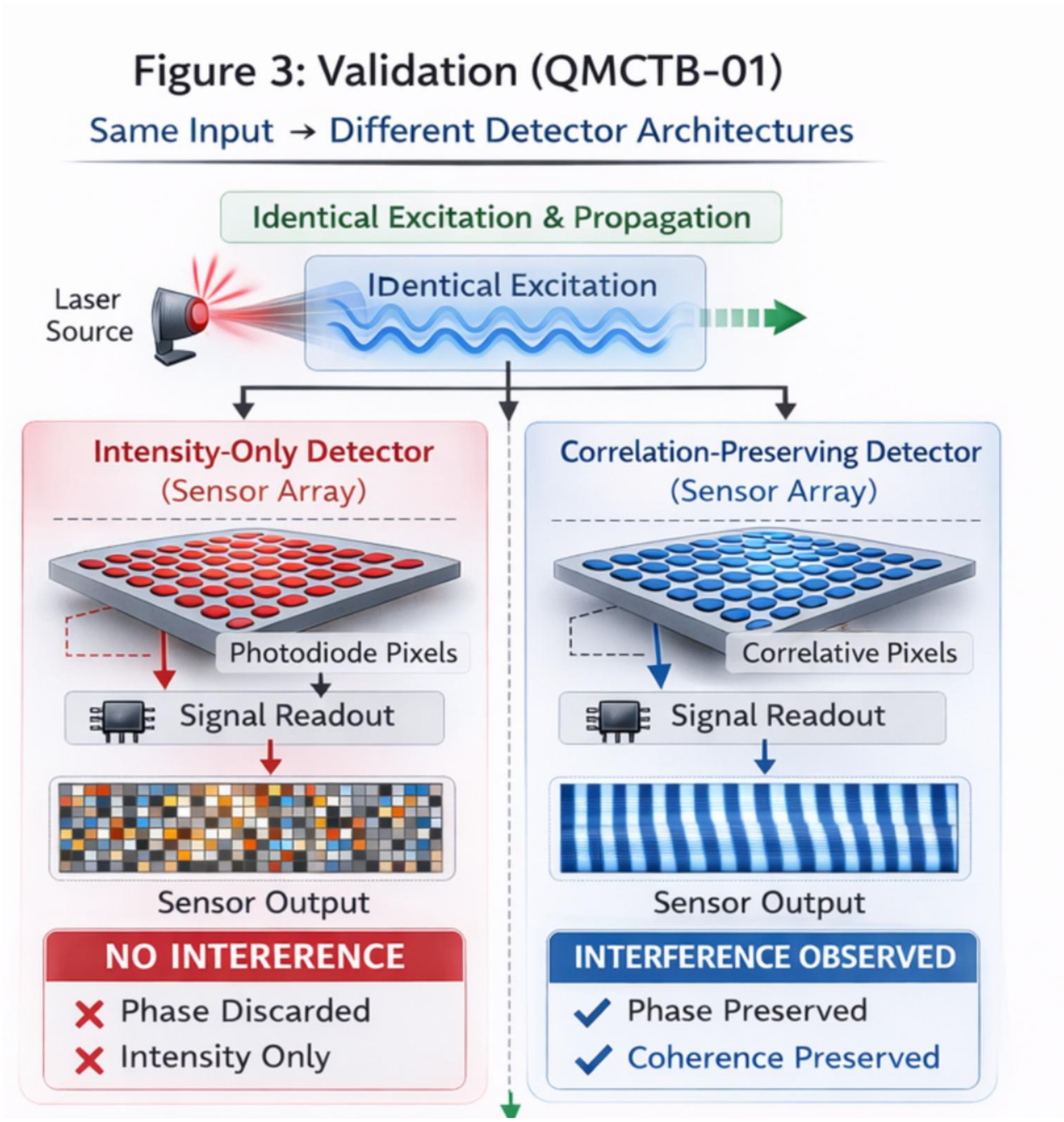
Predicted results:

- Intensity–only detection → no interference (V ≈ 0)

- Correlation–preserving detection → stable interference (V ≈ 0.8—1.0)

Same input → different detector → different observed reality.

Physical instantiation: A near-field double-slit setup with two detector types: (i) intensity-only pixel array, (ii) wavefront-sensitive sensor array preserving inter-pixel phase relationships.




**Same Input → Different Detector → Different Observed Reality**

## 5. Key Insight

**Interference is encoded prior to detection; the detector determines its observability.**

## 6. Conclusion

Detector–Plane Imaging establishes measurement as a detector–defined process operating at the instrumentation layer, where quantum and classical optics converge at the sensor plane. Measurement outcomes are determined by coherence preservation, not photon accumulation.

**Same input → different detector → different observed reality.**

## References

[1] Eisaman, M. D., Fan, J., Migdall, A., & Polyakov, S. V. Single-photon sources and detectors. *Review of Scientific Instruments* **82**, 071101 (2011).

[2] Hadfield, R. H. Single-photon detectors for optical quantum information applications. *Nature Photonics* **3**, 696—705 (2009).

[3] Janesick, J. R. *Scientific Charge-Coupled Devices*. SPIE Press (2001).

[4] Rogalski, A. Infrared detectors: status and trends. *Progress in Quantum Electronics* **27**, 59—210 (2003).

[5] Glauber, R. J. The quantum theory of optical coherence. *Physical Review* **130**, 2529—2539 (1963).

[6] Tonomura, A. et al. Demonstration of single-electron buildup of an interference pattern. *American Journal of Physics* **57**, 117—120 (1989).

[7] Bach, R., Pope, D., Liou, S. H., & Batelaan, H. Controlled double-slit electron diffraction. *New Journal of Physics* **15**, 033018 (2013).

[8] Lemos, G. B. et al. Quantum imaging with undetected photons. *Nature* **512**, 409—412 (2014).

[9] Qian, G. et al. Quantum induced coherence light detection and ranging. *Physical Review Letters* **131**, 033603 (2023).

[10] Aspden, R. S. et al. Heralded quantum imaging. *Optics Express* **23**, 24454—24464 (2015).

[11] Kim, E. M. et al. Quantum optical induced-coherence tomography. *arXiv preprint* arXiv:2309.06777 (2023).

[12] Davenport, D. et al. Quantum ghost imaging microscopy depth-of-field study (2024).

[13] Pearce, E. et al. Practical quantum imaging with undetected photons (2023).

[14] Munshi, S. et al. Near-field imaging of local interference (2025).